NPC – Unit 4 Assignment

Topic: Configure DHCP and Set Up a Network Using Cisco Packet Tracer

***

Name: Sanskriti Santoshkumar Uttam

PRN: 2125UCSF1114

Department : Cybersecurity

Course: Network Protocol Configuration (Fortinet)

Academic Year: 2026–27

---

1. Aim

To implement a network using Cisco Packet Tracer, configure a DHCP server to assign IPv4 addresses automatically to client computers, and test connectivity between the computers using the ping utility.

2. Tools and Software Required

- Cisco Packet Tracer
- One Server-PT device
- One Cisco 2960 switch
- Three PC-PT devices
- Copper straight-through cables

3. Network Design

The network was designed using a star topology. Server0 and the three client computers were connected to a central Cisco 2960 switch. The server provides DHCP services, while the PCs act as clients that request their network configurations automatically.

Since all devices are connected to the same subnet, a router is not required for communication within this network.

Device Connection Details

Device	Interface	Connected Device	Interface
Server0	FastEthernet0	Switch0	FastEthernet0/1
PC0	FastEthernet0	Switch0	FastEthernet0/2
PC1	FastEthernet0	Switch0	FastEthernet0/3
PC2	FastEthernet0	Switch0	FastEthernet0/4

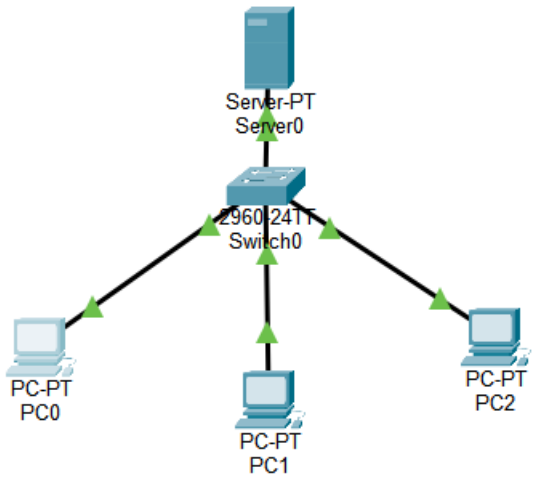


Figure 1: Physical arrangement of the network devices.

4. IPv4 Addressing Scheme

Server0 uses a manually configured IPv4 address so that its DHCP service remains accessible at a consistent address. The client computers obtain their addresses through DHCP.

Network Device	IPv4 Address	Subnet Mask	Address Assignment
Server0	192.168.2.2	255.255.255.0	Manual
PC0	192.168.2.100	255.255.255.0	Automatic
PC1	192.168.2.101	255.255.255.0	Automatic
PC2	192.168.2.102	255.255.255.0	Automatic

The network uses the address range 192.168.2.0/24. Because the setup does

not include a router or a separate DNS server, the default gateway and DNS server values are configured as 0.0.0.0 in the DHCP pool.

5. Implementation Procedure

5.1 Building the Topology

The required network devices were added to the Packet Tracer workspace. Server0, PC0, PC1, and PC2 were connected to separate ports of Switch0 using copper straight-through cables. The connection indicators were checked before proceeding with the configuration.

5.2 Assigning a Static Address to Server0

The server configuration was opened by selecting Server0 and navigating to **Desktop → IP Configuration**. The Static option was selected, and the following parameters were entered.

Parameter	Configured Value
IPv4 Address	192.168.2.2
Subnet Mask	255.255.255.0
Default Gateway	Blank
DNS Server	Blank

5.3 Setting Up the DHCP Pool

The DHCP service was configured from **Server0 → Services → DHCP**. The service was enabled, and a pool named LAN_POOL was created.
The pool parameters are listed below.

DHCP Parameter	Configuration
Pool Name	LAN_POOL
Default Gateway	0.0.0.0
DNS Server	0.0.0.0
Starting IPv4 Address	192.168.2.100
Subnet Mask	255.255.255.0
Maximum Number of Users	50
DHCP Service	On

During the initial setup, the built-in pool named serverPool could not be removed or edited through the controls available at that time. As a result, PC0 initially received the address 192.168.2.1. The configuration was then adjusted by setting serverPool to start at 192.168.2.100 with a maximum of 50 users, and the changes were saved.

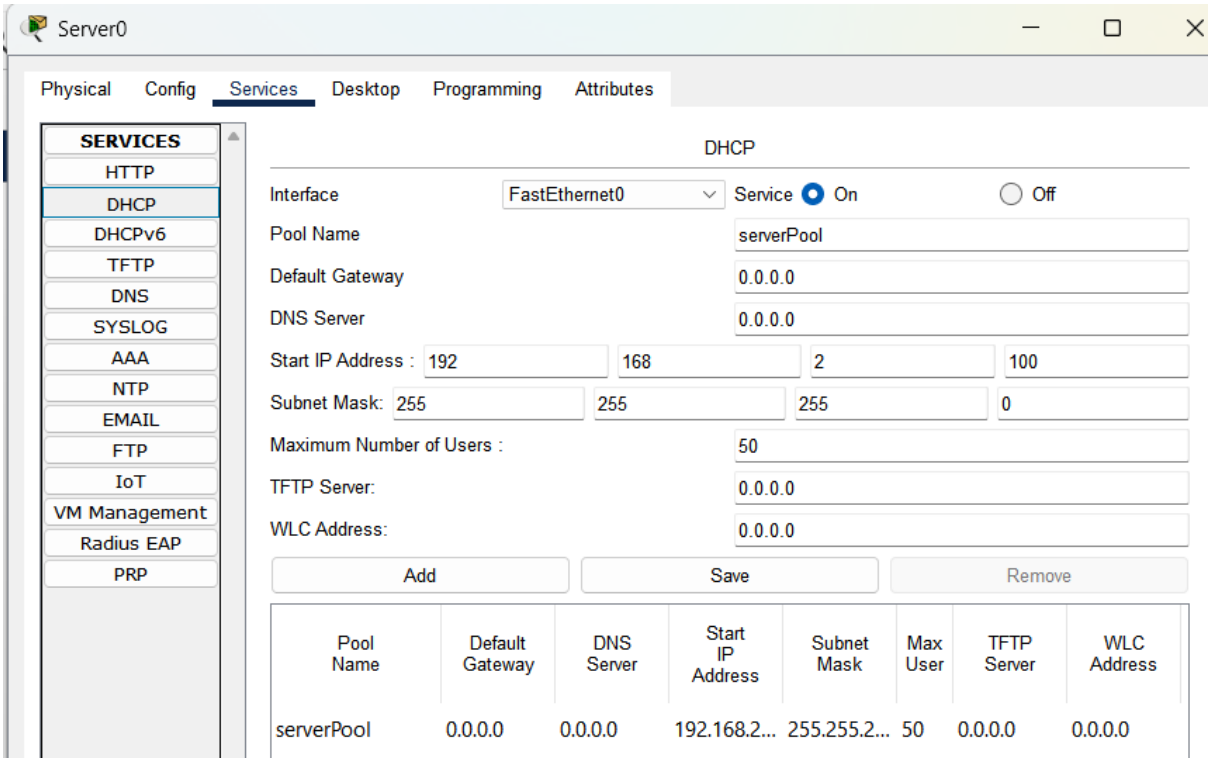


Figure 2: DHCP service settings in Server0.

5.4 Enabling DHCP on the Client Computers

Each client was configured to request its network settings from the DHCP server.

The following steps were completed on each PC:

1. Open the PC configuration window.
2. Navigate to **Desktop → IP Configuration**.
3. Select **DHCP** instead of Static.
4. Wait for the DHCP request to complete.
5. Check the displayed IPv4 address and subnet mask.

The expected address allocation is shown below.

Client	Assigned IPv4 Address	Subnet Mask
PC0	192.168.2.100	255.255.255.0

PC1	192.168.2.101	255.255.255.0
PC2	192.168.2.102	255.255.255.0

The message “DHCP request successful” indicates that the client successfully obtained its configuration.

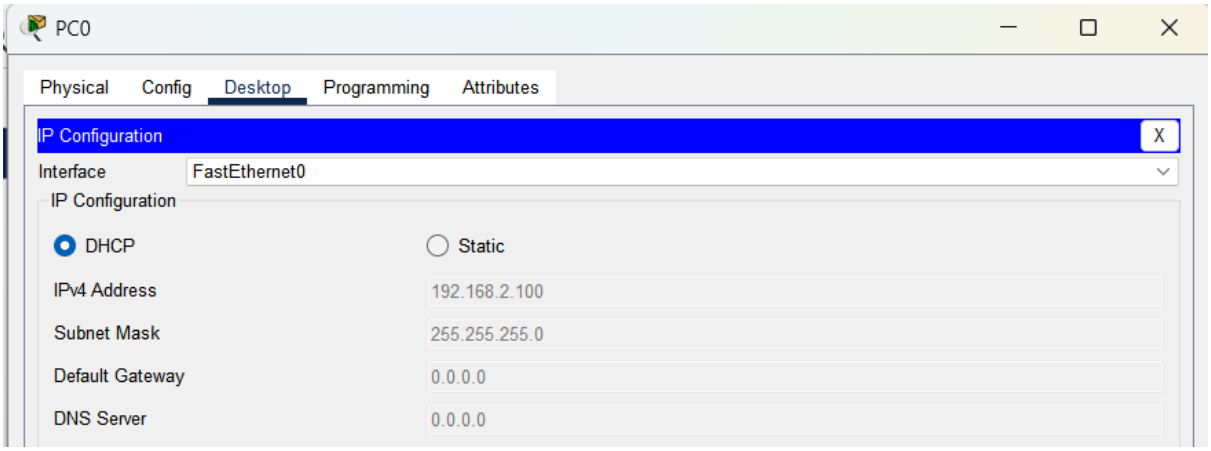


Figure 3: Successful DHCP configuration on a client PC.

5.5 Checking the Assigned Addresses

The Command Prompt on each client computer was used to inspect the current network configuration. The following command was executed:

ipconfig

The command displays the IPv4 address and subnet mask assigned to the device. The addresses were checked to ensure that the clients had unique addresses within the same subnet.

6. Network Connectivity Verification

After assigning the addresses, communication between the client computers was tested from PC0. The ping utility was used to send packets to PC1 and PC2.

The commands executed were:

ping 192.168.2.101

ping 192.168.2.102

The first command checks connectivity with PC1, while the second checks connectivity with PC2. Successful replies indicate that the destination devices are reachable over the local network.

7. Observations and Test Results

The results of the DHCP and connectivity checks are summarized in the following table.

Verification Test	Observation	Status
DHCP allocation on PC0	Address 192.168.2.100 assigned	Successful
DHCP allocation on PC1	Address 192.168.2.101 assigned	Successful
DHCP allocation on PC2	Address 192.168.2.102 assigned	Successful
Ping from PC0 to PC1	Four replies received; 0% packet loss	Successful
Ping from PC0 to PC2	Four replies received; 0% packet loss	Successful

The connectivity tests showed that PC0 could reach both client computers. All four packets were received in each ping test, with no packet loss.

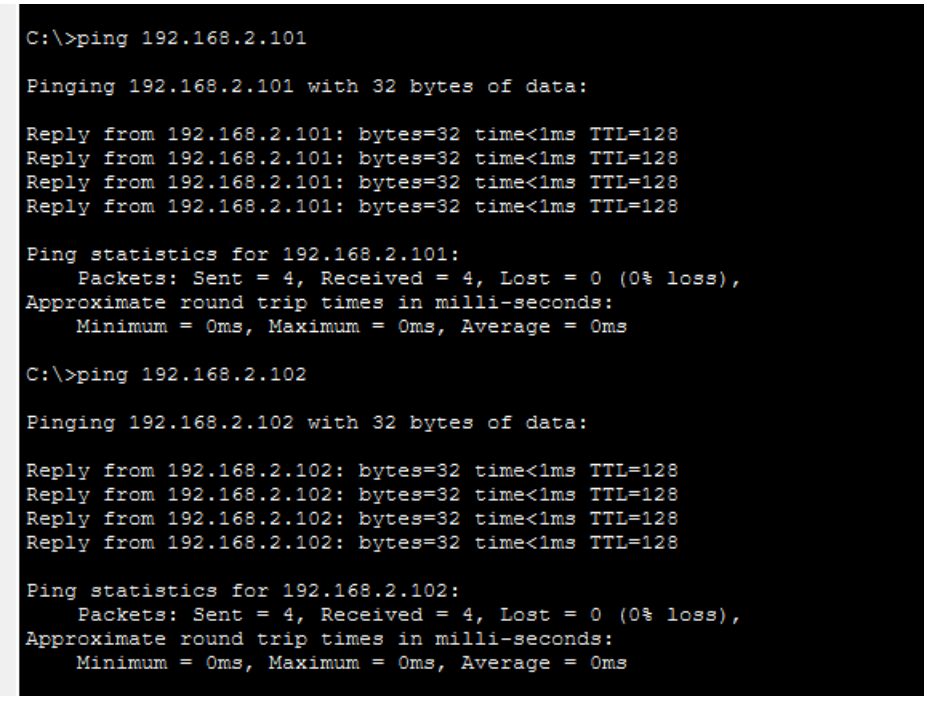


Figure 4: Connectivity verification between the client computers.

8. Discussion

The experiment demonstrates the role of DHCP in simplifying network configuration. Instead of entering an IP address and subnet mask manually on every client, the computers can request these settings from the DHCP server. The initial address assignment issue also highlighted the importance of checking the active DHCP pool configuration. After the pool settings were

adjusted, the client addresses and connectivity results matched the intended network configuration.

9. Result

The DHCP server was configured in Cisco Packet Tracer, and the three client PCs obtained the intended IPv4 addresses automatically:

- PC0 — 192.168.2.100
- PC1 — 192.168.2.101
- PC2 — 192.168.2.102

Connectivity tests from PC0 to PC1 and PC2 each returned four successful replies with 0% packet loss.

10. Conclusion

The experiment successfully demonstrated automatic IP address allocation using DHCP in a local area network. A server was configured to provide network settings to client computers connected through a switch. The assigned addresses were verified using ipconfig, and connectivity was tested using ping. The results illustrate how DHCP reduces manual configuration effort and helps organize IP address allocation. The successful connectivity tests confirmed communication between the tested devices within the same subnet.